

EDUCATOR'S GUIDE

VibeX Expansion

Practical Usage Guide for Classroom and Workshop Settings

AI Literacy · Sustainability Education · Digital Ethics

AUDIENCE

University students
Introductory AI literacy workshops
Sustainability & digital ethics courses

SESSION

30–45 minutes total
Individual or groups of 2–4
10–30 classroom participants

For facilitation enquiries, contact your course coordinator.

SESSION DURATION

Activity	Duration
Introduction to AI infrastructure & sustainability	5–10 min
Gameplay session	10–15 min
Reflection and discussion	10–15 min
<i>Total</i>	<i>30–45 min</i>

EXAMPLE CLASSROOM FLOW

Step 1 Introduction & Context Setting

5–10 min

The facilitator introduces the topic of AI infrastructure and explains that modern AI systems require significant amounts of electricity, water, computing hardware, and physical datacenter infrastructure. Examples may include:

- ◇ Cooling systems used in AI datacenters
- ◇ Electricity usage of large AI models
- ◇ Growth in AI demand worldwide
- ◇ Environmental trade-offs of rapid AI expansion

The facilitator also explains the game objective and how player decisions influence AI growth and surrounding communities.

Step 2 Gameplay Session

10–15 min

Students play *VibeX Expansion* individually or in small groups, acting as decision-makers responsible for expanding AI centers while balancing economic growth, environmental sustainability, and public wellbeing. Decisions include:

- ◇ AI center expansion
- ◇ Infrastructure placement
- ◇ Resource allocation
- ◇ Community impact
- ◇ Energy and water usage

Different choices lead to different outcomes, strategies may prioritise rapid growth or environmental protection.

Step 3 Observation & In-Game Reflection

Students observe how their choices affect:

- ◇ Environmental sustainability
- ◇ Water and energy use
- ◇ Village wellbeing
- ◇ Infrastructure demand
- ◇ AI service growth
- ◇ Public satisfaction

Students are encouraged to compare strategies and reflect on the consequences of different decision-making approaches.

Step 4 Post-Game Discussion

10–15 min

Facilitators guide a discussion in which students reflect on:

- ◇ Which strategy they followed and why they made certain decisions
- ◇ The trade-offs they encountered along the way
- ◇ Whether their choices were sustainable in the long term
- ◇ What surprised or challenged them during gameplay

SUGGESTED DISCUSSION TOPICS

Energy Consumption of AI Systems How much electricity modern AI systems and datacenters require, and why energy demand increases as AI adoption grows.

Water Usage in AI Datacenters Why datacenters require cooling systems and how water consumption can impact surrounding environments and communities.

Hardware & Infrastructure Requirements The physical infrastructure behind AI systems, including servers, cooling facilities, networking equipment, and land usage.

Ethical Trade-offs in AI Expansion The balance between technological progress, economic growth, environmental sustainability, and public wellbeing.

Responsible Prompting Habits How thoughtful and efficient prompting can reduce unnecessary AI usage and resource consumption.

Long-Term Sustainability of AI Whether current AI growth patterns are environmentally sustainable and what future improvements may be necessary.

LEARNING OUTCOMES

AFTER COMPLETING THIS ACTIVITY, STUDENTS SHOULD BE ABLE TO:

- ◇ Describe the environmental costs of AI systems
- ◇ Understand the infrastructure required to support AI
- ◇ Recognise trade-offs between AI growth and sustainability
- ◇ Reflect critically on their own AI usage habits
- ◇ Discuss responsible and environmentally conscious AI practices

TECHNICAL REQUIREMENTS

- ◇ Modern web browser
- ◇ Laptop or desktop computer
- ◇ Mouse recommended
- ◇ Stable internet connection
(if hosted online)

COURSE INTEGRATION

- ◇ AI Literacy
- ◇ Sustainability Education
- ◇ Digital Responsibility
- ◇ Technology Ethics
- ◇ Human–Computer Interaction
- ◇ Introduction to Artificial Intelligence